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AI/ML intern

Raritone Summer Internship 2026

Project Report

1. INTRODUCTION

The rapid growth of e-commerce has transformed the fashion industry by providing customers with convenient access to a wide variety of clothing and accessories. However, one of the major limitations of online fashion shopping is the inability of customers to physically try products before making a purchase. This often leads to uncertainty regarding garment fit, appearance, and style suitability, resulting in reduced customer confidence and increased product returns.

Artificial Intelligence (AI), Machine Learning (ML), and Computer Vision technologies have emerged as powerful solutions to address these challenges. Modern fashion platforms are increasingly adopting AI-powered systems that can analyze user images, estimate body measurements, detect human poses, segment garments, and generate personalized fashion recommendations. These technologies enable customers to visualize how clothing may appear on them before purchase, thereby enhancing the overall shopping experience.

The primary focus of this project was to explore various AI technologies that contribute to intelligent fashion applications. Throughout the internship, research was conducted on human body detection, pose estimation, body measurement extraction, cloth segmentation, virtual try-on systems, recommendation engines, and AR/VR-based fashion experiences. Different AI models, frameworks, and datasets were studied to understand how these technologies can be integrated into a unified fashion platform.

Special attention was given to 2D and 3D Virtual Try-On systems, which represent one of the most promising applications of AI in the fashion industry. These systems combine pose detection, segmentation, garment alignment, and simulation techniques to create realistic visualizations of clothing on users. In addition, AI-powered recommendation systems were explored to understand how personalized outfit suggestions can improve customer satisfaction and engagement.

This project provides an overview of the technologies, models, datasets, workflows, and research findings related to AI-driven fashion solutions. The knowledge gained through this study demonstrates the potential of Artificial Intelligence in creating smarter, more interactive, and highly personalized fashion shopping experiences.

2. OBJECTIVES

The primary objective of this project was to explore and understand the application of Artificial Intelligence, Machine Learning, and Computer Vision technologies in the fashion industry. The project focused on studying modern AI-powered solutions that can improve online shopping

experiences through virtual try-on systems, body analysis, and personalized fashion recommendations.

The specific objectives of the project were:

- To study human body detection techniques and understand how AI models identify body landmarks and key points.
- To explore human pose estimation methods and analyze their role in virtual try-on and body measurement systems.
- To understand body measurement extraction techniques using image-based analysis.
- To research cloth segmentation and human parsing models used for garment detection and clothing extraction.
- To investigate 2D Virtual Try-On systems and understand garment alignment, garment warping, and fitting workflows.
- To explore 3D Virtual Try-On technologies, including avatar generation, body mesh reconstruction, and cloth simulation.
- To study AI-powered fashion recommendation systems and understand different recommendation approaches.
- To collect and analyze datasets used for pose detection, segmentation, virtual try-on, and fashion recommendation research.
- To compare various AI models and frameworks used in fashion technology applications.
- To design a complete AI workflow demonstrating the integration of body detection, segmentation, virtual try-on, and recommendation systems.
- To identify challenges associated with AI-powered fashion systems and explore possible solutions and future improvements.

Through these objectives, the project aimed to gain a comprehensive understanding of how Artificial Intelligence can be utilized to create intelligent, personalized, and interactive fashion shopping experiences.

3. RESEARCH AREAS COVERED

During the internship, extensive research was conducted on various Artificial Intelligence, Machine Learning, and Computer Vision technologies used in modern fashion applications. The objective was to understand how different AI techniques can be integrated to create intelligent virtual try-on systems and personalized fashion recommendation platforms.

3.1 Human Body Detection

Human body detection forms the foundation of many fashion technology applications. Research was conducted on methods used to identify human body structures from images and videos. Body detection systems help locate users within an image and provide the necessary information required for measurement estimation, pose analysis, and garment fitting.

The study focused on understanding body landmark detection and how body keypoints can be used to support virtual try-on systems and recommendation engines.

3.2 Human Pose Estimation

Pose estimation is the process of detecting and tracking key body joints and landmarks. Research was conducted on pose estimation techniques that enable AI systems to understand human posture and movement.

Technologies such as MediaPipe and OpenPose were explored to understand how body keypoints are extracted and how pose information contributes to virtual try-on applications. Accurate pose estimation plays an important role in garment alignment and fitting accuracy.

3.3 Body Measurement Analysis

Research was conducted on image-based body measurement estimation techniques. The study focused on understanding how measurements such as shoulder width, body proportions, waist estimation, and height estimation can be derived using pose landmarks.

Body measurement analysis is essential for improving garment fitting and providing personalized fashion experiences.

3.4 Cloth Segmentation and Human Parsing

Cloth segmentation is a critical component of virtual try-on systems. Research was performed on segmentation models that separate clothing from the background and identify different garment regions.

Human parsing and segmentation techniques were studied to understand how AI systems generate clothing masks and extract garments from images. This process helps improve garment placement and overall virtual try-on quality.

3.5 2D Virtual Try-On Systems

A major area of research focused on 2D Virtual Try-On technologies. The complete workflow of virtual fitting systems was analyzed, including image preprocessing, pose detection, segmentation, garment alignment, and try-on generation.

Research was conducted on garment warping techniques and fitting methods that help generate realistic clothing visualizations on user images.

3.6 3D Virtual Try-On Technologies

The internship also included research on 3D virtual try-on systems. Topics explored included 3D avatar generation, body mesh reconstruction, cloth simulation, and motion tracking.

The study highlighted how 3D technologies can provide more realistic garment visualization and immersive shopping experiences compared to traditional 2D systems.

3.7 AI Fashion Recommendation Systems

Research was conducted on recommendation systems used in fashion platforms. Different approaches such as content-based filtering, collaborative filtering, and hybrid recommendation systems were analyzed.

The objective was to understand how AI can provide personalized outfit suggestions based on user preferences, shopping behavior, body characteristics, and fashion trends.

3.8 AR/VR Fashion Technologies

Emerging technologies such as Augmented Reality (AR) and Virtual Reality (VR) were explored as part of the project. Research focused on AR-based clothing visualization, virtual fitting rooms, and immersive fashion experiences.

The study demonstrated how AR and VR can enhance customer engagement and improve online shopping interactions.

3.9 Dataset Research and Analysis

Various datasets commonly used in fashion AI applications were researched and analyzed. Datasets such as DeepFashion, DeepFashion2, and VITON-HD were studied to understand their role in training and evaluating AI models.

Dataset research provided insights into data collection, organization, and the importance of diverse training data for improving AI performance.

Overall, these research areas provided a comprehensive understanding of the technologies, models, and workflows involved in AI-powered fashion systems and virtual try-on applications.

4. **MODELS EXPLORED**

Various Artificial Intelligence and Computer Vision models were explored during the internship to understand their applications in body analysis, segmentation, virtual try-on systems, and recommendation engines. These models play an important role in developing intelligent fashion technology solutions.

4.1 MediaPipe

MediaPipe is a widely used framework developed for real-time perception tasks. It provides efficient body landmark detection and pose estimation capabilities.

During the research, MediaPipe was studied for:

- Human pose detection
- Body landmark extraction
- Skeleton tracking
- Body measurement estimation

Advantages:

- Fast processing speed
- Real-time performance
- Easy integration with computer vision applications

MediaPipe is particularly useful for virtual try-on systems where body landmarks are required for clothing alignment and measurement estimation.

4.2 OpenPose

OpenPose is a popular pose estimation framework capable of detecting multiple body keypoints from images and videos.

Research focused on:

- Human pose estimation
- Multi-person pose detection
- Body keypoint tracking
- Motion analysis

Advantages:

- High pose estimation accuracy
- Detailed body landmark detection
- Supports full-body analysis

OpenPose provides valuable pose information that can improve garment fitting and virtual try-on accuracy.

4.3 U2Net

U2Net is a deep learning model commonly used for image segmentation and background removal.

Research applications included:

- Cloth segmentation
- Garment extraction
- Human image masking
- Background removal

Advantages:

- High-quality segmentation masks
- Accurate object separation
- Effective clothing extraction

U2Net plays an important role in preparing images for virtual try-on workflows.

4.4 Detectron2

Detectron2 is a powerful object detection and segmentation framework developed for advanced computer vision tasks.

Research focused on:

- Human parsing
- Instance segmentation
- Garment detection
- Clothing region identification

Advantages:

- Detailed segmentation results
- Strong object recognition capabilities
- Flexible architecture

Detectron2 is useful for identifying clothing regions and generating segmentation outputs required for virtual fitting systems.

4.5 HR-VITON

HR-VITON is a modern virtual try-on architecture designed to generate realistic clothing fitting results.

Research areas included:

- Garment warping
- Clothing alignment
- Virtual try-on workflows
- Realistic garment fitting

Advantages:

- Improved fitting quality

- Better clothing preservation
- Enhanced visual realism

HR-VITON demonstrates how advanced AI models can create realistic virtual try-on experiences for fashion applications.

4.6 Recommendation Models

Different recommendation approaches were explored to understand how personalized fashion suggestions can be generated.

Content-Based Filtering:
Recommends products based on item characteristics and user interests.

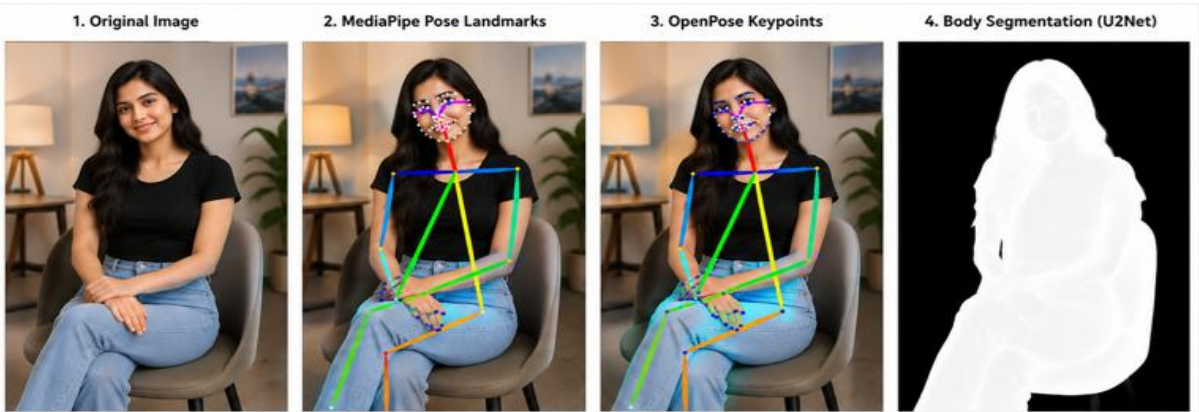
Collaborative Filtering:
Provides recommendations based on the behavior and preferences of similar users.

Hybrid Recommendation Systems:
Combine multiple recommendation techniques to improve recommendation accuracy and personalization.

- Advantages:
- Personalized user experience
 - Better product discovery
 - Increased customer engagement

Model Comparison Summary

Model	Primary Application
MediaPipe	Pose Detection & Landmark Tracking
OpenPose	Human Pose Estimation
U2Net	Cloth Segmentation
Detectron2	Human Parsing & Segmentation
HR-VITON	Virtual Try-On
Hybrid Recommendation Models	Personalized Fashion Recommendations





The exploration of these models provided a comprehensive understanding of the technologies used in modern AI-powered fashion systems and virtual try-on platforms.

5. DATASETS COLLECTED

Datasets play a crucial role in the development and evaluation of Artificial Intelligence models. High-quality datasets enable AI systems to learn patterns, improve prediction accuracy, and generate reliable outputs. During the internship, several datasets related to fashion technology, pose estimation, segmentation, and virtual try-on systems were researched and analyzed.

5.1 DeepFashion

DeepFashion is one of the most widely used datasets in fashion-related research. It contains a large collection of clothing images along with detailed annotations describing garment categories, attributes, and fashion styles.

Applications:

- Fashion recognition
- Clothing classification
- Attribute prediction
- Recommendation system research

Advantages:

- Large-scale dataset
- Rich clothing annotations
- Diverse fashion categories

DeepFashion provides valuable data for understanding clothing characteristics and fashion trends.

5.2 DeepFashion2

DeepFashion2 is an advanced version of the DeepFashion dataset that includes additional annotations and supports more complex computer vision tasks.

Applications:

- Garment detection
- Clothing segmentation

- Human parsing
- Fashion image analysis

Advantages:

- Detailed garment annotations
- Diverse clothing categories
- Multiple viewpoints and poses

DeepFashion2 is particularly useful for segmentation and virtual try-on research because of its detailed clothing information.

5.3 VITON-HD Dataset

VITON-HD is a dataset specifically designed for virtual try-on research. It contains images of people and garments that can be used to evaluate virtual fitting systems.

Applications:

- Virtual try-on systems
- Garment alignment
- Garment warping research
- Clothing fitting analysis

Advantages:

- High-resolution images
- Suitable for virtual fitting experiments
- Supports realistic try-on research

The dataset helps researchers evaluate how accurately garments can be aligned and fitted onto user images.

5.4 Human Pose Datasets

Human pose datasets contain images annotated with body landmarks and keypoints. These datasets are commonly used for training and evaluating pose estimation models.

Applications:

- Human pose estimation
- Body landmark detection
- Skeleton tracking
- Motion analysis

Advantages:

- Detailed body keypoint annotations
- Useful for pose estimation research
- Supports body measurement analysis

Pose datasets play an important role in understanding body structure and movement, which are essential components of virtual try-on systems.

5.5 Body Segmentation Datasets

Body segmentation datasets are used to train AI models that separate body parts and clothing regions from images.

Applications:

- Human parsing
- Cloth segmentation
- Image masking
- Virtual try-on preprocessing

Advantages:

- Improved segmentation accuracy
- Better clothing extraction
- Enhanced virtual fitting quality

These datasets contribute to creating accurate segmentation masks required for realistic garment visualization.

Importance of Datasets in Fashion AI

The research conducted during the internship highlighted the importance of high-quality datasets in developing reliable AI systems. Diverse datasets help improve model performance, support multiple body types and clothing styles, and enhance the realism of virtual try-on outputs.

Dataset Summary

Dataset	Purpose
DeepFashion	Fashion Recognition & Classification
DeepFashion2	Segmentation & Garment Detection
VITON-HD	Virtual Try-On Research
Human Pose Datasets	Pose Estimation

Body Segmentation Datasets Human Parsing & Segmentation

Overall, the study of these datasets provided a deeper understanding of how data contributes to the development of AI-powered fashion technologies and personalized shopping experiences.

6. 2D VIRTUAL TRY-ON RESEARCH & FINDINGS

One of the primary focus areas of this internship was the study and analysis of 2D Virtual Try-On systems. Virtual Try-On technology enables users to visualize how garments may appear on their bodies before making a purchase. This technology aims to improve customer confidence, enhance shopping experiences, and reduce product return rates.

A 2D Virtual Try-On system combines several Artificial Intelligence and Computer Vision techniques to generate realistic clothing visualizations. The workflow generally involves body detection, pose estimation, segmentation, garment alignment, and image generation.

2D Virtual Try-On Workflow

The typical workflow followed in a 2D Virtual Try-On system includes:

User Image
↓
Body Detection
↓
Pose Estimation
↓
Body Segmentation
↓
Garment Alignment
↓
Virtual Try-On Output

Each stage contributes to generating an accurate and realistic representation of how a garment would appear on a user.

Body Detection and Pose Estimation

The first stage involves identifying the human body and extracting body landmarks. Technologies such as MediaPipe and OpenPose are commonly used to detect key body points and estimate posture.

Pose information is essential because clothing must be aligned according to the user's body position. Accurate pose estimation significantly improves the quality of the final try-on output.

Body Segmentation

After detecting the body, segmentation techniques are used to separate the user from the background and identify different body regions. Segmentation models generate masks that help isolate clothing and body areas.

Accurate segmentation is important because errors at this stage can negatively affect garment placement and fitting quality. Research on U2Net and Detectron2 demonstrated how segmentation models contribute to realistic virtual fitting experiences.

Garment Alignment and Warping

One of the most important components of virtual try-on systems is garment alignment. Since users may have different body shapes and poses, garments must be transformed and adjusted before being placed on the user's image.

Garment warping techniques allow clothing to adapt to body structure while maintaining garment appearance. Research on HR-VITON and similar virtual try-on architectures highlighted the importance of garment warping in improving realism and fitting accuracy.

Image Preprocessing

Image preprocessing is another important step in the virtual try-on pipeline. Activities such as image resizing, background handling, and quality enhancement help improve overall system performance.

Proper preprocessing ensures that AI models receive cleaner inputs and generate more reliable outputs.

Key Findings

The research conducted on 2D Virtual Try-On systems resulted in several important observations:

- Accurate pose detection improves garment placement and fitting quality.
- Effective body segmentation enhances the realism of generated outputs.

- Garment alignment and warping are critical for achieving natural-looking results.
- Image preprocessing contributes to improved output quality and model performance.
- High-quality datasets are essential for training reliable virtual try-on models.
- The combination of pose estimation, segmentation, and garment fitting techniques creates more realistic user experiences.

Benefits of 2D Virtual Try-On Systems

2D Virtual Try-On technology offers several advantages for both customers and fashion retailers:

- Enhanced online shopping experience.
- Better visualization of clothing before purchase.
- Increased customer confidence.
- Reduced uncertainty regarding garment fit.
- Potential reduction in product return rates.
- Improved customer engagement.

Overall, the study of 2D Virtual Try-On systems provided valuable insights into how Artificial Intelligence and Computer Vision technologies can transform the online fashion industry by creating more personalized and interactive shopping experiences.

7. 3D VIRTUAL TRY-ON RESEARCH & FINDINGS

In addition to 2D Virtual Try-On systems, research was conducted on 3D Virtual Try-On technologies to understand how advanced fashion platforms provide immersive and realistic shopping experiences. Unlike 2D systems, which overlay garments onto images, 3D Virtual Try-On systems create digital representations of users and simulate clothing behavior in a three-dimensional environment.

The primary objective of 3D Virtual Try-On systems is to allow users to visualize garments from different angles while accurately representing body shape, size, and movement.

Key Research Areas

3D Avatar Generation

A major component of 3D Virtual Try-On systems is avatar generation. Research was conducted on how user images and body measurements can be used to create digital avatars that closely resemble the user.

These avatars serve as virtual representations of customers and provide a foundation for realistic garment visualization.

Body Mesh Reconstruction

Body mesh reconstruction involves creating a three-dimensional model of the human body. This process enables AI systems to estimate body shape and proportions more accurately.

Research highlighted the importance of body mesh models in improving garment fitting and creating realistic virtual fitting experiences.

Cloth Simulation

Cloth simulation is responsible for modeling how garments behave when worn by a user. It helps represent factors such as fabric movement, stretching, folding, and draping.

Realistic cloth simulation enhances the visual quality of virtual try-on systems and provides users with a better understanding of garment appearance.

Motion Tracking

Motion tracking technologies allow virtual avatars to replicate user movements. This capability improves realism by enabling garments to respond dynamically to body motion.

Motion tracking is particularly useful in AR and VR environments where real-time interaction is required.

Tools and Technologies Explored

Several tools and frameworks commonly used in 3D fashion applications were researched:

- Blender – 3D modeling and visualization.
- Unity – Interactive 3D application development.
- Three.js – Web-based 3D rendering.
- SMPL Models – Human body modeling and reconstruction.

These technologies are widely used for developing realistic virtual fitting experiences and interactive fashion applications.

Key Findings

The research conducted on 3D Virtual Try-On systems resulted in several important observations:

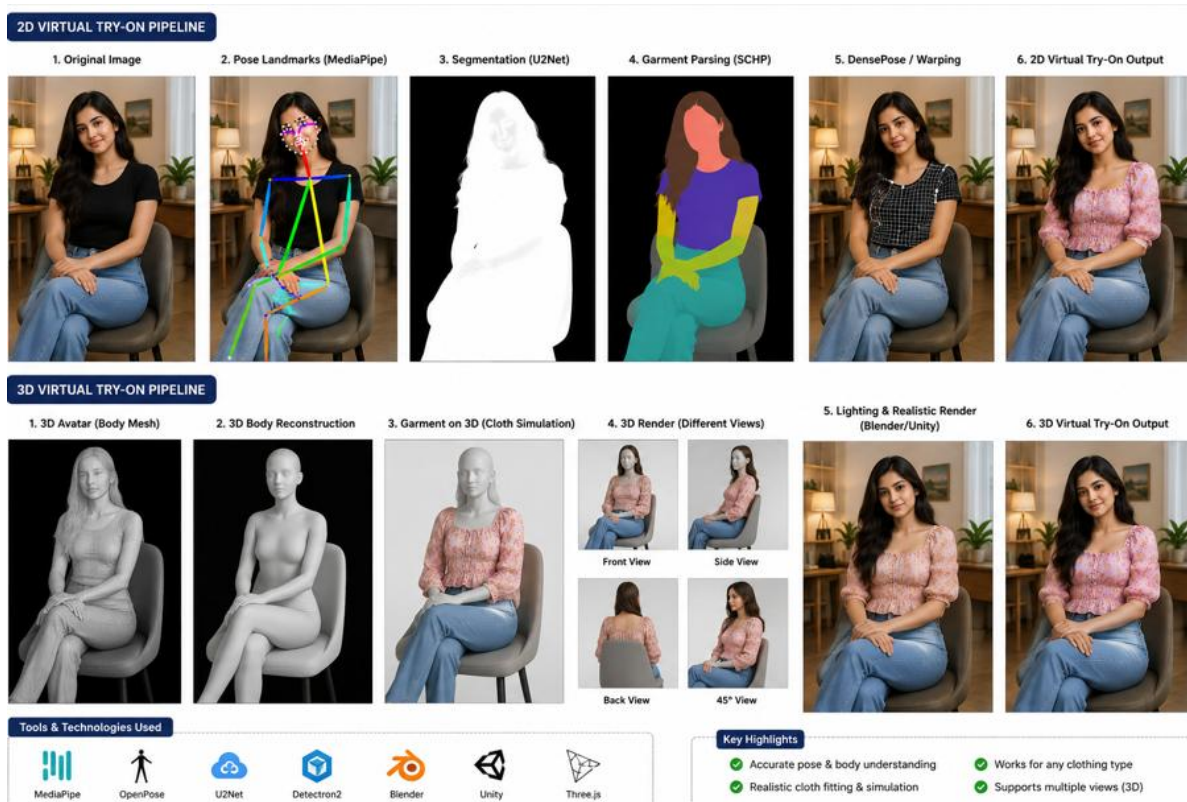
- 3D avatars provide a more realistic representation of users.
- Body mesh reconstruction improves garment fitting accuracy.
- Cloth simulation enhances realism by mimicking fabric behavior.
- Motion tracking creates interactive and engaging experiences.
- 3D systems offer a more immersive shopping experience compared to traditional 2D approaches.
- Advanced computational resources are required to support real-time rendering and simulation.

Benefits of 3D Virtual Try-On Systems

The study revealed several advantages of adopting 3D Virtual Try-On technologies:

- Improved visualization of garments.
- Enhanced customer confidence before purchase.
- More realistic fitting experiences.
- Greater user engagement.
- Reduced uncertainty regarding clothing size and appearance.
- Potential integration with AR and VR shopping environments.

Overall, 3D Virtual Try-On technologies represent the next stage in the evolution of digital fashion experiences. Although these systems are more complex than traditional 2D approaches, they provide significant opportunities for creating highly immersive and personalized online shopping platforms.



8. AI FASHION RECOMMENDATION SYSTEM

Artificial Intelligence has transformed the way fashion products are recommended to customers. Modern e-commerce platforms use recommendation systems to analyze user preferences and provide personalized suggestions that improve customer engagement and shopping experiences.

During the internship, research was conducted on various recommendation approaches used in fashion technology platforms. The objective was to understand how AI can assist users in discovering products that match their preferences, body type, and fashion interests.

Importance of Fashion Recommendation Systems

Online fashion platforms often contain thousands of products, making it difficult for users to find items that suit their needs. Recommendation systems help simplify this process by suggesting relevant products based on user behavior and product characteristics.

Personalized recommendations provide several benefits:

- Improved customer experience.
- Faster product discovery.
- Increased customer engagement.
- Better shopping satisfaction.
- Enhanced business performance.

Recommendation Approaches Studied

Content-Based Filtering

Content-Based Filtering recommends products based on their attributes and similarities. The system analyzes characteristics such as clothing type, color, style, fabric, and category to suggest similar products.

For example, if a user frequently views casual shirts, the system may recommend other shirts with similar features.

Advantages:

- Personalized recommendations.
- Easy implementation.
- Effective for new products.

Collaborative Filtering

Collaborative Filtering generates recommendations based on the behavior and preferences of other users. The system identifies users with similar interests and recommends products that those users have interacted with.

For example, if multiple users with similar preferences purchased certain clothing items, those products may be recommended to new users with similar behavior.

Advantages:

- Learns from user interactions.
- Provides diverse recommendations.
- Discovers hidden user interests.

Hybrid Recommendation Systems

Hybrid Recommendation Systems combine multiple recommendation techniques to improve recommendation quality and accuracy.

By integrating Content-Based and Collaborative Filtering approaches, hybrid systems can overcome the limitations of individual methods and provide more relevant recommendations.

Advantages:

- Higher recommendation accuracy.
- Better personalization.
- Improved user satisfaction.
- Enhanced product discovery.

Fashion Recommendation Factors

The research identified several factors that can influence recommendation quality:

- User Preferences
- Body Type
- Clothing Style
- Fashion Trends
- Purchase History
- Browsing Behavior
- Seasonal Preferences

Incorporating these factors allows recommendation systems to provide more meaningful and personalized fashion suggestions.

Future Opportunities

Future recommendation systems can become more intelligent by integrating body measurement analysis, virtual try-on results, and fashion trend prediction models. Such systems could recommend outfits that not only match user preferences but also suit their body shape and fitting requirements.

Key Findings

The study of AI Fashion Recommendation Systems resulted in the following observations:

- Personalized recommendations improve user engagement.
- Hybrid approaches provide better recommendation accuracy.
- User behavior plays an important role in recommendation quality.
- AI can significantly enhance product discovery and customer satisfaction.
- Recommendation systems can complement virtual try-on technologies to create a complete intelligent fashion platform.

Overall, AI-powered recommendation systems represent a valuable component of modern fashion technology applications. When combined with virtual try-on systems, they can deliver highly personalized and interactive shopping experiences that benefit both customers and businesses.

9. COMPLETE AI WORKFLOW

A major objective of this project was to understand how different Artificial Intelligence and Computer Vision technologies can be integrated into a single workflow to create an intelligent fashion platform. Throughout the internship, various modules including body detection, pose estimation, segmentation, virtual try-on systems, and recommendation engines were studied and analyzed.

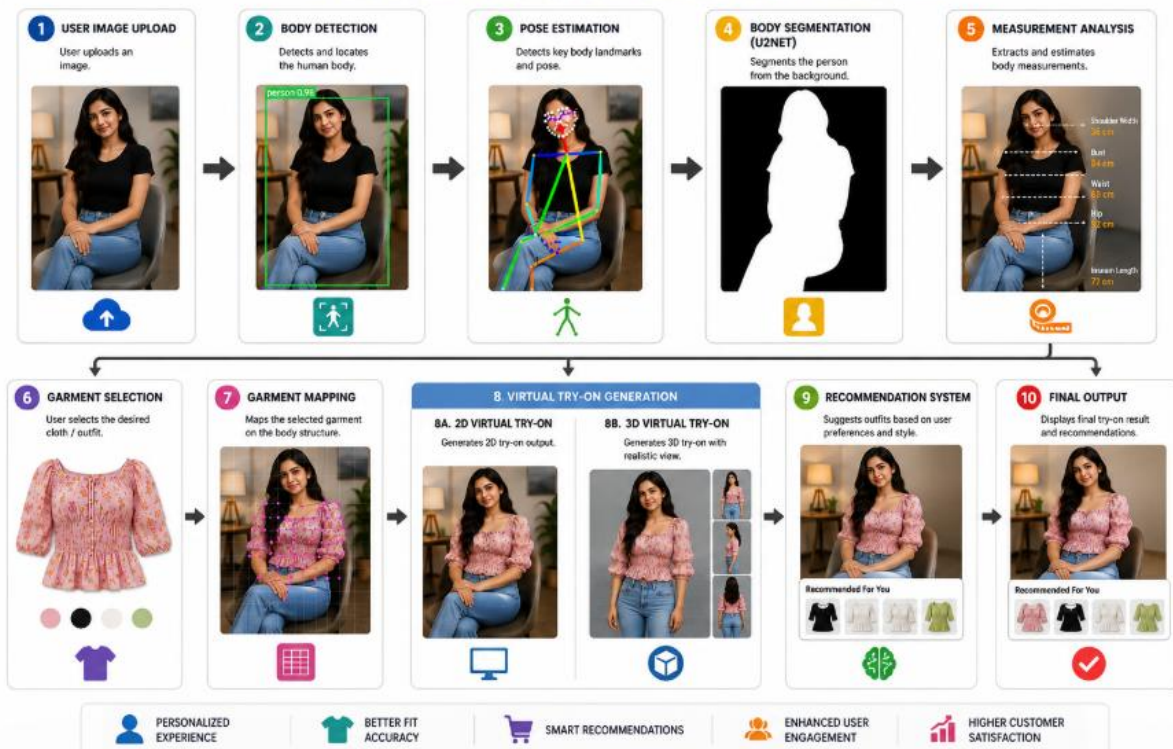
The complete AI workflow demonstrates how these individual components work together to provide a seamless and personalized fashion shopping experience.

AI Workflow

```
graph TD;
    A[User Upload] --> B[Body Detection];
    B --> C[Pose Estimation];
    C --> D[Body Segmentation];
    D --> E[Measurement Analysis];
    E --> F[Garment Mapping];
    F --> G[Virtual Try-On];
    G --> H[Fashion Recommendation];
    H --> I[Final Output];
```

User Upload
↓
Body Detection
↓
Pose Estimation
↓
Body Segmentation
↓
Measurement Analysis
↓
Garment Mapping
↓
Virtual Try-On
↓
Fashion Recommendation
↓
Final Output

AI FASHION PLATFORM – COMPLETE WORKFLOW



1. User Upload

The workflow begins when a user uploads an image to the platform. This image serves as the primary input for subsequent AI processing stages.

The uploaded image contains information about the user's body structure, pose, and appearance, which is required for virtual fitting and recommendation generation.

2. Body Detection

The first AI processing stage involves identifying the human body within the image. Body detection systems locate the user and extract relevant body regions required for further analysis.

This step ensures that the system can accurately focus on the user rather than background elements.

3. Pose Estimation

After body detection, pose estimation algorithms identify body landmarks and keypoints. These landmarks represent important body joints and positions.

Pose information helps determine the user's posture and enables accurate garment alignment during virtual try-on generation.

4. Body Segmentation

Body segmentation separates the user from the background and identifies different body and clothing regions.

This process generates segmentation masks that improve garment placement accuracy and contribute to more realistic virtual fitting results.

5. Measurement Analysis

Using body landmarks and segmentation outputs, measurement estimation techniques can be applied to determine body proportions and approximate measurements.

Measurement analysis supports personalization and improves fitting accuracy.

6. Garment Mapping

At this stage, selected garments are aligned with the user's body structure and pose information.

Garment mapping ensures that clothing is positioned correctly before virtual try-on generation.

7. Virtual Try-On Generation

Virtual try-on systems combine body information, segmentation outputs, and garment mapping techniques to generate a realistic visualization of how clothing may appear on the user.

This stage represents one of the most important components of the overall workflow.

8. Fashion Recommendation

After generating virtual try-on results, recommendation systems can analyze user preferences, body characteristics, and fashion trends to provide personalized outfit suggestions.

This integration helps users discover products that best match their interests and requirements.

9. Final Output

The final output includes virtual try-on visualizations and personalized fashion recommendations.

This creates an interactive shopping experience where users can visualize garments, explore recommendations, and make more informed purchasing decisions.

Workflow Significance

The integration of multiple AI modules into a unified workflow demonstrates how Artificial Intelligence can transform online fashion retail. Each stage contributes to improving personalization, visualization quality, recommendation accuracy, and overall user experience.

The complete workflow represents a foundation for future AI-powered fashion platforms capable of delivering intelligent, interactive, and customer-centric shopping experiences.

10. FUTURE SCOPE

The rapid advancement of Artificial Intelligence, Computer Vision, and Augmented Reality technologies presents significant opportunities for the future development of intelligent fashion platforms. The research conducted during this internship highlighted several areas where virtual try-on systems and recommendation engines can be further improved to provide more personalized, realistic, and interactive shopping experiences.

1. Real-Time Virtual Try-On

Future virtual try-on systems can move beyond static image processing and provide real-time fitting experiences using live camera feeds. Users would be able to instantly visualize garments on themselves while moving naturally, creating a more engaging shopping environment.

Benefits:

- Improved user interaction
- Enhanced shopping experience
- Increased customer confidence
- Reduced product return rates

2. Advanced 3D Virtual Try-On

The integration of advanced 3D technologies can significantly improve virtual fitting realism. Future systems can utilize highly detailed avatars, body mesh reconstruction, and realistic cloth simulation to create accurate representations of garment appearance and fit.

Benefits:

- More realistic garment visualization
- Better fitting accuracy
- Enhanced customer satisfaction
- Improved purchase decisions

3. AI-Powered Personal Stylist

Future recommendation systems can evolve into intelligent personal stylists capable of understanding user preferences, body type, fashion trends, and purchase history.

These systems can provide personalized outfit suggestions, styling advice, and wardrobe recommendations.

Benefits:

- Personalized shopping experiences
- Improved customer engagement
- Better product discovery
- Increased user retention

4. AR-Based Fashion Shopping

Augmented Reality technology has the potential to transform online shopping by enabling users to virtually try clothing, accessories, footwear, and jewelry through mobile devices.

Future AR-based solutions can bridge the gap between physical and digital shopping experiences.

Benefits:

- Interactive product visualization
- Greater customer engagement
- Improved purchasing confidence
- Innovative shopping experiences

5. Improved Body Measurement Estimation

Future research can focus on developing more accurate body measurement extraction techniques using advanced pose estimation and computer vision models.

More precise measurements can improve garment fitting recommendations and enhance virtual try-on quality.

Benefits:

- Better size recommendations
- Improved fitting accuracy

- Increased customer trust
- Reduced size-related returns

6. Enhanced Recommendation Systems

Recommendation systems can be improved by incorporating additional factors such as social media trends, seasonal preferences, user behavior analytics, and fashion forecasting.

Future hybrid recommendation models can provide highly personalized and context-aware fashion suggestions.

Benefits:

- More relevant recommendations
- Improved personalization
- Enhanced customer satisfaction
- Increased sales opportunities

7. Mobile Application Integration

As mobile commerce continues to grow, integrating AI-powered virtual try-on and recommendation systems into mobile applications can improve accessibility and convenience.

Mobile-based solutions can provide users with intelligent fashion assistance anytime and anywhere.

Benefits:

- Greater accessibility
- Improved user convenience
- Expanded platform reach
- Enhanced customer experience

8. Scalable AI Infrastructure

Future platforms will require scalable AI architectures capable of handling large volumes of users and data efficiently. Optimized AI pipelines and cloud-based deployments can support real-time processing while maintaining high performance.

Benefits:

- Faster processing
- Better scalability
- Improved reliability
- Enhanced platform performance

Overall, the future of AI-powered fashion technology is highly promising. Continued advancements in Artificial Intelligence, Computer Vision, AR/VR, and recommendation systems will enable fashion platforms to provide increasingly personalized, immersive, and intelligent shopping experiences.

11. CONCLUSION

The Raritone Summer Internship 2026 provided a valuable opportunity to explore the applications of Artificial Intelligence, Machine Learning, and Computer Vision in the fashion industry. Throughout the internship, extensive research was conducted on technologies that have the potential to transform online shopping experiences through personalization, automation, and intelligent decision-making.

The project focused on understanding key components of AI-powered fashion systems, including human body detection, pose estimation, body measurement analysis, cloth segmentation, virtual try-on technologies, recommendation systems, and AR/VR-based fashion experiences. Various AI models such as MediaPipe, OpenPose, U2Net, Detectron2, and HR-VITON were studied to understand their functionality and applications in modern fashion platforms.

Research on datasets including DeepFashion, DeepFashion2, and VITON-HD provided insights into the importance of high-quality data for training and evaluating AI systems. The exploration of both 2D and 3D Virtual Try-On technologies demonstrated how advanced AI techniques can improve garment visualization, fitting accuracy, and customer engagement.

The study of recommendation systems highlighted the significance of personalization in fashion e-commerce. Content-based filtering, collaborative filtering, and hybrid recommendation approaches were analyzed to understand how AI can provide relevant outfit suggestions based on user preferences and behavior.

A complete AI workflow was also designed to demonstrate how body detection, pose estimation, segmentation, garment mapping, virtual try-on generation, and recommendation systems can be integrated into a unified fashion platform. This workflow illustrates the potential of combining multiple AI technologies to create intelligent and interactive shopping experiences.

In addition to technical knowledge, the internship enhanced important professional skills such as teamwork, research methodology, documentation, communication, problem-solving, and project coordination. Working on a collaborative project helped strengthen the ability to analyze emerging technologies and present findings in a structured manner.

Overall, this internship provided a comprehensive understanding of AI-driven fashion technologies and their real-world applications. The knowledge and experience gained through this project will serve as a strong foundation for future academic work, research activities, and professional opportunities in the fields of Artificial Intelligence, Machine Learning, Computer Vision, and Fashion Technology.

The project successfully achieved its objectives and demonstrated how AI can play a significant role in creating smarter, more personalized, and engaging fashion experiences for users in the future.